# UNIT 1: OPERATING SYSTEMS

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| Content: | 1. Alexis Leon and Mathews Leon, "Fundamentals of Information Technology", Vikas Publication, Second edition, 2009. | No of hours |
|---|---|---|
| Unit - 1 | **Introduction to Computers:** Importance of computers, characteristics of computers, classification of computers, uses of computers.<br><br>**Anatomy of Digital Computer:** parts of computer, CPU: Control Unit and ALU. secondary storage devices, keyboards, mouse, scanners, readers, digital cameras, monitors, and printers.<br><br>**Operating Systems:** Introduction, functions of an operating system, classification of operating systems.<br><br>**Introduction to Computer Problem Solving:** Introduction, problem-solving aspect, top-down design. | 10 |

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- An **operating system** manages and coordinates **the functions performed by the computer hardware including the CPU, input/output devices, secondary storage devices and communication and network equipment**.
- It performs **basic tasks** such as **recognizing input from the keyboard, sending output to the display screen,** keeping track of files and directories on the disk and **controlling peripheral devices** such as disk drives and printers.
- It **keeps track of each hardware resource, determine who gets what resource, when the user will have access to the resource, terminate access** etc.
- Primary purpose of the operating system is to **maximize the productivity of a computer** system by operating it in the most efficient manner and **minimizing the amount of human intervention required.**
- Operating systems must **be loaded and activated before accomplishing other tasks**. This emphasizes that operating systems is the most indispensable component of the software interface between users and the hardware.
- **Examples**: Windows, Linux, MacOS etc.

# FUNCTIONS OF AN OPERATING SYSTEM

1. **Job management:** It **recognizes** the jobs, **identifies** their **priorities**, **determines** whether appropriate **memory is available**, **schedules** and **finally runs each job** at the appropriate moment.
2. **Batch processing: Data** are accumulated and **processed in groups**.
3. **On-line processing:** Data are **processed instantaneously**. Example: A sales person may need to find out whether a particular item requested by customer is in stock for immediate shipment. Using online system the request for information will be instantly acknowledged by the online software.
4. **Data Management:** Manage **storage and retrieval of data.**
5. **Virtual Storage:** It is possible to **increase the capacity of main memory without actually increasing the size**. This is done by breaking a job into pages and keeping only few of these in main memory. As a result large jobs can be processed by a CPU.
6. **Input/Output management:** Operating system also **manage the input to and output from the computer system**. This applies to the flow of data among computers , terminals and other devices such as printers.

## CLASSIFICATION OF OPERATING SYSTEMS

- **One way of classifying operating systems** is based on their **use**-desktop, server, mainframe etc.
- Another way is **based on capabilities and features-** multiuser, multiprocessing, multitasking etc

**1. Desktop operating systems**

- Run on desktop computers.
- Generally OS in this category include windows, Mac OS, Linux etc.
- Windows is currently the leading desktop operating system.

**2. Server operating system**

- It **provides services, resources, and functionalities** to multiple users or **client devices across a network.**
- **Server OS are optimized for tasks such as managing databases, running web services, ensuring high levels of security etc.**
- Some common examples of server operating systems include:

❖ Windows Server
❖ UNIX-based systems
❖ macOS Server

**3. Mainframe operating systems**

- **Mainframe computers are powerful, large-scale computers** primarily used by large organizations for critical applications, such as **bulk data processing, large-scale transaction processing, high-volume input/output operations**.
- **Mainframe operating systems** are specialized OS **designed to manage and run the powerful and complex hardware of mainframe computers.**
- These operating systems are **optimized for high-volume, high-reliability, and high-availability environments**, typically found in large organizations that require extensive data processing capabilities.

**Common Mainframe Operating Systems:**

- ❖ **z/OS :Developer**: IBM
- ❖ **z/VM**: **Developer**: IBM

### 4. Multi-user operating systems

•**Simultaneous Access**: Multiple users can log in and work on the system at the same time, often from different locations or terminals.

•**Resource Management**: The **operating system efficiently manages hardware resources, ensuring fair allocation among all users**. This includes CPU time, memory, disk space, and input/output devices. **Examples**: Windows server, UNIX

### 5. Multiprocessing operating system

- A **multiprocessing operating system** is designed to support and manage **multiple processors (or CPUs)** within a single computer system.
- This allows the **system to execute multiple processes simultaneously**, significantly improving performance and efficiency, especially for tasks that can be parallelized.

**Key Characteristics:**

❖ **Multiple CPUs**: The system can have two or more processors (CPUs) that share tasks.

❖ **Parallel Processing**: Tasks **can be divided into smaller sub-tasks**, which are then executed **in parallel** across multiple processors, **leading to faster execution**.

## 4. Multi-tasking operating systems

- Multitasking is the ability to **execute more than one task(program) at the same time**.
- CPU **switches** from **one program to another quickly**.

**There are two basic types of multitasking**

- ❖ **Preemptive multitasking:** the operating system can interrupt a running process, pause it, and allocate CPU resources to another process.
- ❖ **Cooperative multitasking:** each program can control the CPU for as long as it needs it.

## 5. Multithreading operating system

- Allows **different parts of a single program to run concurrently**.
- It executes **different parts of a program called threads** simultaneously.

## 6. Real time operating systems

**Real-Time Operating Systems (RTOS)** are designed to handle tasks with precise timing constraints and provide guaranteed response times for critical operations.

**Applications of Real-Time Operating Systems:**

- **Aerospace and Defense**: satellite systems, and military applications
- **Automotive Systems**:Managing vehicle control systems
- **Medical Devices**:Controlling medical equipment

- Examples:
- CCP (communication control programs)
- iRMX( Real time multitasking executive for Intel)

# THANK YOU☺

by PRACHI SURLAKAR

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